

NOVACARE HEALTH INSURANCE

Insurance Claims Data Cleaning Report

Records Processed: 6,150

Columns: 19 → 27 (post-cleaning)

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1. Executive Summary

This report documents the end-to-end data quality assessment and cleaning process applied to the NovaCare Health Systems insurance claims dataset. The raw dataset contained 6,150 claim records across 19 columns covering patient demographics, provider information, clinical codes, financial figures, and claim processing metadata.

Data quality issues were identified across five major categories: inconsistent text casing and formatting, missing values in financial and categorical fields, duplicate claim entries, date logic violations, and erroneous numerical values. A structured SQL-based cleaning pipeline was applied, resulting in a standardized, audit-ready claims_clean table with eight additional derived and flag columns for downstream analysis.

6,150 Total Claims Records processed	19 Original Columns Input schema	27 Final Columns After enrichment
480 Duplicate Claims is_duplicate_claim = 1	121 Date Logic Errors Submission before discharge	123 Negative Processing Days Corrected to absolute value
74 Paid Anomalies Paid with no billed/allowed amt	183 Denied, Unknown Reason Reason tagged as Unknown	553 Pending Claims Flagged for follow-up

2. Dataset Overview

2.1 Source Data

The source file novacare_claims_raw contained 6,150 rows and 19 columns. Each row represents a single insurance claim submitted on behalf of a patient encounter. The table below describes each column, its data type, and its role in the dataset.

Column Name	Type	Description
claim_id	VARCHAR	Unique identifier for each claim (Primary Key)
patient_id	VARCHAR	Identifier for the patient associated with the claim
encounter_id	VARCHAR	Identifier for the clinical encounter — should be unique per visit
provider_id	VARCHAR	Identifier for the treating provider
provider_name	VARCHAR	Full name of the treating provider (had casing inconsistencies)
department	VARCHAR	Clinical department (e.g., Cardiology, Emergency) — had many variant names
admission_date	DATE	Date patient was admitted
discharge_date	DATE	Date patient was discharged
claim_submission_date	DATE	Date the claim was submitted to the payer
icd10_primary	VARCHAR	Primary ICD-10 diagnosis code (had embedded periods)
icd10_description	VARCHAR	Plain-text description of the ICD-10 code
cpt_procedure	INT	CPT procedure code for the primary procedure

		performed
billed_amount	FLOAT	Amount billed by the provider (188 nulls)
allowed_amount	FLOAT	Payer-approved amount (188 nulls)
paid_amount	FLOAT	Amount actually paid — no nulls but had anomalies
claim_status	VARCHAR	Status of the claim: Paid/Denied/Pending/Appealed (had 12 casing variants)
denial_reason	VARCHAR	Reason for denial if applicable (3,620 nulls — mostly non-denied claims)
payer_type	VARCHAR	Insurance/Government/Self-Pay/NULL (249 nulls)
claim_processing_days	INT	Days taken to process the claim (123 negative values)

3. Data Quality Issues Identified

3.1 Text Standardization Issues

Several text columns contained multiple representations of the same value, resulting from inconsistent data entry across systems or time periods.

3.1.1 Provider Name (provider_name)

The provider_name field contained at least six different formatting styles for each of the 12 providers. Examples found in the raw data include:

Format Type	Raw Example	Cleaned To
All caps, no title	AMARA OSEI	Dr. Amara Osei

All caps with prefix	DR AMARA OSEI	Dr. Amara Osei
Lowercase	amara osei	Dr. Amara Osei
Inverted surname-first	Osei, Amara Dr.	Dr. Amara Osei
Correct format	Dr. Amara Osei	Dr. Amara Osei

Resolution: Provider names were standardized to the canonical 'Dr. [First] [Last]' format using a CASE statement keyed on provider_id, ensuring each of the 12 providers maps to exactly one name format.

3.1.2 Department (department)

The department column had 33 distinct raw values representing only 8 unique departments. Variants included abbreviated forms, all-caps entries, and mixed-case entries.

Department	Raw Variants Found	Standardized To
Cardiology	CARDIOLOGY, Cardio, cardiology	Cardiology
Emergency	EMERGENCY, ER, Emerg, emergency	Emergency
General Surgery	GENERAL SURGERY, Gen Surgery, Gen. Surgery	General Surgery
Internal Medicine	INTERNAL MEDICINE, Int Medicine, Int. Med	Internal Medicine
Neurology	NEUROLOGY, Neuro, neurology	Neurology
Oncology	ONCOLOGY, Onco, oncology	Oncology
Orthopedics	ORTHOPEDICS, Ortho, orthopedics	Orthopedics

Pediatrics	PEDIATRICS, Peds, pediatrics	Pediatrics
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Note: One residual variant 'Gen. Surgery' was still observed in the cleaned output for a small number of records, indicating this variant may not have been fully captured by the UPPER() matching logic. This is noted as a remaining data quality gap.

3.1.3 Claim Status (claim_status)

The claim_status field had 12 unique raw values representing only 4 valid statuses: Paid, Denied, Pending, and Appealed. The inconsistency was purely a casing issue (PAID, paid, Paid, DENIED, denied, Denied, etc.).

Resolution: A CASE WHEN UPPER(TRIM(claim_status)) = 'PAID' THEN 'Paid' ... statement normalized all statuses to title case. Post-cleaning, only 4 values exist, consolidating what was a fragmented field.

Raw Status Values	Count	Clean Status Values	Count
Paid / PAID / paid	2,884 (combined)	Paid	2,884
Denied / DENIED / denied	2,434 (combined)	Denied	2,434
Pending / PENDING / pending	553 (combined)	Pending	553
Appealed / APPEALED / appealed	279 (combined)	Appealed	279

3.1.4 ICD-10 Codes (icd10_primary)

ICD-10 codes in the raw dataset contained embedded periods (e.g., N39.0, I25.10, M54.5), which is non-standard for database storage and can break joins or code lookups. All 15 affected diagnosis codes had their periods stripped to yield clean codes (e.g., N390, I2510, M545).

3.2 Missing Values

Three columns contained NULL values in the raw dataset. Each was handled differently depending on the clinical context of the field.

Column	Raw Nulls	Clean Nulls	Resolution Applied
billed_amount	188	188*	NULL retained; no imputation applied. Noted for downstream review.
allowed_amount	188	188*	Logic: COALESCE(allowed_amount, paid_amount) to fill from paid_amount where available; remaining NULLs set to 0.
denial_reason	3,620	3,437	Denied claims with NULL reason tagged as 'Unknown' (183 records). Non-denied NULLs are expected and retained.
payer_type	249	249*	SQL UPDATE to 'Unknown' was scripted but NULL count remains 249 — a potential execution gap.

** Discrepancy noted: The SQL scripts include UPDATE statements to replace NULL billed_amount with 0 and NULL payer_type with 'Unknown'. However, the exported cleaned CSV still shows 188 NULL billed_amount and 249 NULL payer_type values. These may not have been applied before export and should be re-verified in the live database.*

3.3 Duplicate Encounter Records

The encounter_id field, which should uniquely identify a patient visit, was found to have 150 duplicate encounter IDs in the raw dataset affecting 480 claim records. Upon investigation, these duplicates carried 'Duplicate Claim' as their denial reason, indicating they were already identified in the source system.

Resolution: A new binary column is_duplicate_claim was added. Records where denial_reason = 'DUPLICATE CLAIM' were flagged with a value of 1 (480 records). These claims are retained in the dataset for audit traceability but should be excluded from revenue and reimbursement analytics.

3.4 Financial Anomalies

3.4.1 Paid Amount Without Billing Records (paid_error)

74 claim records had NULL billed_amount and NULL allowed_amount, yet carried a positive paid_amount value with a claim status of 'Paid'. This is a financial integrity violation — a claim cannot be paid without a billed and approved amount on record.

Resolution: These records were flagged with paid_error = 1 for investigation. No values were altered to preserve the audit trail.

3.4.2 Financial Metrics Summary

The core financial columns showed no change in distribution between raw and clean, confirming that the cleaning process did not alter valid financial data.

Metric	Billed Amount	Allowed Amount	Paid Amount
Records with Value	5,962	5,962	6,150

Mean	\$3,066.72	\$2,373.82	\$1,111.22
Median	\$2,801.76	\$2,147.53	\$0.00
Minimum	\$172.16	\$108.72	\$0.00
Maximum	\$8,714.06	\$7,880.07	\$7,727.10

3.5 Date Logic Violations

3.5.1 Submission Before Discharge (date_error)

121 records had a claim_submission_date that preceded the discharge_date. This is logically impossible in a standard claims workflow — a claim cannot be filed before the patient has been discharged from care.

Resolution: These records were flagged with date_error = 1. The SQL script also attempted to flag submissions before the admission date, though due to a variable naming error in the script (claim_before_discharge was updated instead of claim_before_admission), this second check may not have executed correctly. This is flagged for re-review.

3.5.2 Days to Submit (days_to_submit)

A derived column days_to_submit was calculated as claim_submission_date minus discharge_date, representing how many days elapsed between discharge and claim filing. The distribution is summarized below.

Metric	Value	Interpretation
Mean	22.7 days	Average time from discharge to claim submission
Median	23 days	Typical filing lag

Minimum	-10 days	Negative values confirm date logic errors still in field
Maximum	56 days	Maximum lag observed — may risk timely filing limits
25th Percentile	11 days	Faster-submitting claims
75th Percentile	34 days	Slower-submitting claims

3.6 Negative Claim Processing Days

123 records contained negative values in the `claim_processing_days` column, which records the number of days taken by the payer to process a claim. Negative processing days are operationally impossible.

Resolution: The `ABS()` function was applied to convert all negative values to their positive equivalents. The 123 affected records were additionally flagged via the `negative_claim_processing_days` column (value = 1) for audit tracking. The underlying cause of negative entries — likely a date subtraction order error in the source system — is recommended for upstream investigation.

4. Derived & Flag Columns Added

Eight new columns were added to the `claims_clean` table. These columns support operational monitoring, audit trail preservation, and downstream analytics without modifying the original source fields.

Column Added	Purpose	Count Flagged
<code>is_duplicate_claim</code>	Binary flag (1/0). Marks claims	480 (7.8%)

	identified as duplicates based on denial_reason = 'Duplicate Claim'. Should be excluded from financial reporting.	
paid_error	Binary flag (1/0). Marks claims with paid_amount > 0 but NULL billed_amount and NULL allowed_amount — a financial integrity violation.	74 (1.2%)
date_error	Binary flag (1/0). Marks claims where claim_submission_date < discharge_date — logically invalid submission timing.	121 (2.0%)
claim_before_discharge	Binary flag (1/0). Underlying column for date_error. Also re-used (incorrectly) for submission-before-admission check due to a naming bug in the SQL.	121 (2.0%)
days_to_submit	Computed integer. Measures days between discharge_date and claim_submission_date. Negative values present where date_error = 1.	6,150 rows
is_pending	Binary flag (1/0). Marks all claims with claim_status = 'Pending' for operational follow-up workflows.	553 (9.0%)
negative_claim_processing_days	Binary flag (1/0). Marks records where claim_processing_days was originally negative before	123 (2.0%)

	correction via ABS().	
length_of_stay_days	Computed integer. Calculates patient length of stay as discharge_date minus admission_date. Range: 1-10 days, mean 5.4 days.	6,150 rows

5. Overall Cleaning Summary

The table below consolidates all data quality checks and their resolution status across both the raw and cleaned datasets.

Issue / Check	Raw Dataset	Clean Dataset	Status
Whitespace trimming (all text fields)	Present in raw	Resolved	Resolved
Provider name casing/format (12 providers)	~60 raw variants	12 canonical values	Resolved
Department casing/abbreviation (8 depts)	33 raw variants	8 + 1 residual	Partial
Claim status casing (4 statuses)	12 raw variants	4 canonical values	Resolved
ICD-10 code periods stripped	Codes had periods	Periods removed	Resolved
NULL billed_amount	188 nulls	188 nulls (not zeroed)	Partial
NULL allowed_amount	188 nulls	188 nulls (not zeroed)	Partial

NULL payer_type	249 nulls	249 nulls (not replaced)	Partial
NULL denial_reason (denied claims)	183 nulls	Tagged 'Unknown'	Resolved
Duplicate encounter_id claims	480 records	Flagged (not removed)	Flagged
Paid with no billing records	74 records	Flagged (not altered)	Flagged
Claim submitted before discharge	121 records	Flagged (not altered)	Flagged
Claim submitted before admission	Not quantified	SQL bug — not fully applied	Partial
Negative claim_processing_days	123 records	ABS() applied + flagged	Resolved
Pending claims tracking	No flag existed	is_pending column added	Resolved

6. Recommendations & Next Steps

6.1 Immediate Actions

- Re-run the NULL replacement UPDATE statements for billed_amount (set to 0) and payer_type (set to 'Unknown') and verify via COUNT query before re-exporting the cleaned dataset.
- Fix the SQL naming bug in the claim_before_admission block: UPDATE claims_clean SET claim_before_admission = 1 WHERE claim_submission_date < admission_date. The current script incorrectly updates claim_before_discharge instead.

- Resolve the 'Gen. Surgery' residual variant in the department column by adding an explicit WHEN clause for it in the standardization CASE statement.

6.2 Analytical Exclusions

- Always filter `is_duplicate_claim = 0` before running revenue, denial rate, or payer analytics to avoid double-counting.
- Exclude `paid_error = 1` records from reimbursement rate calculations until the financial anomaly is resolved.
- Use `date_error = 0` as a baseline filter for any time-to-submit or filing lag analyses.

6.3 Upstream / Process Improvements

- Investigate the source system that produces negative `claim_processing_days` values — this likely reflects a date-field order error in the ETL or source transaction system.
- Add a NOT NULL constraint on `payer_type` at the source ingestion layer to prevent future NULL entries.
- Enforce ICD-10 code formatting (no periods) as a validation rule at point of entry or during ETL transformation.
- Implement provider name as a lookup/reference table relationship rather than a free-text field to eliminate casing inconsistencies at the source.
- Consider adding a `claim_filing_deadline` field to support timely filing compliance monitoring, given that some claims are submitted up to 56 days post-discharge.

7. Appendix — Denial Reason Distribution

The following breakdown of denial reasons in the cleaned dataset is provided for reference. Records where the claim was not denied carry NULL in this field by design.

Denial Reason	Count	% of Denied Claims
Duplicate Claim	480	19.7%
Medical Necessity	368	15.1%
Coding Error	349	14.3%
Patient Not Eligible	346	14.2%
Incomplete Documentation	338	13.9%
Timely Filing Exceeded	333	13.7%
Missing Authorization	316	13.0%
Unknown (was NULL)	183	7.5%
Not Denied (NULL — expected)	3,437	N/A

End of Report — NovaCare Health Systems Claims Data Cleaning Report | April 2025